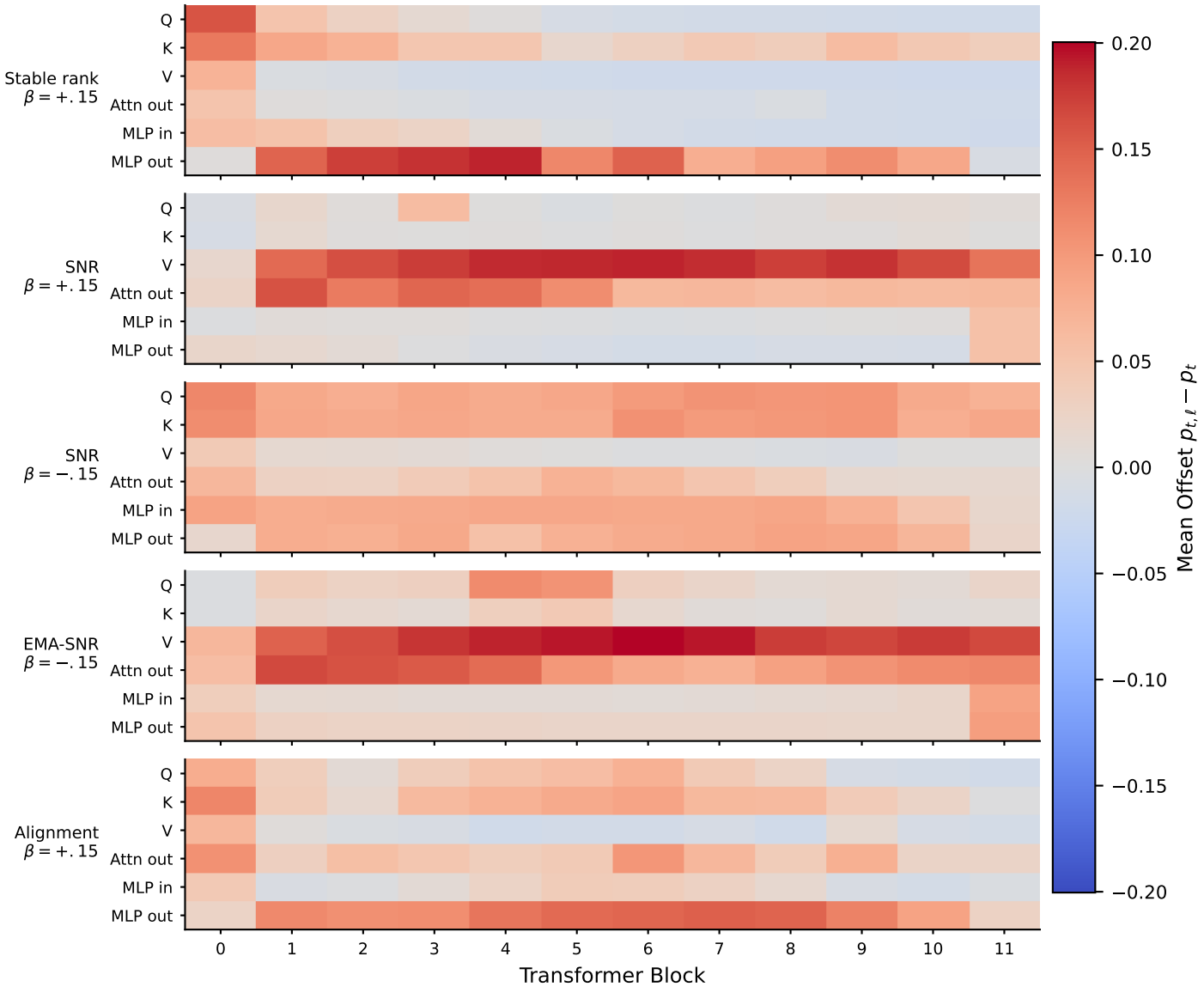

Proxy Routers Change Different Matrix Types at Different Depths
Stable rank
$\beta = +.15$
SNR
$\beta = +.15$
SNR
$\beta = -.15$
EMA-SNR
$\beta = -.15$
Alignment
$\beta = +.15$
Q
K
V
Attn out
MLP in
MLP out
0
1
2
3
4
5
6
7
8
9
10
11
Transformer Block
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Mean Offset $p_{t,\ell} - p_t$